

Madhavanesh K

Software Development Engineer | Blockchain Developer | Full Stack Developer

<http://linkedin.com/in/madhavanesh-kannan-8b7192203> | +918610259198 | madhavanesh@gmail.com | Coimbatore, India

Software Development Engineer with 1.7 years of experience in **full-stack web development** and **blockchain solutions** at scale. Proven in working with cross-functional teams, gathering requirements, and executing project tasks efficiently for **startups**.

SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python.
- **Frontend Development:** React.js, Next.js, HTML5, CSS3, Tailwind CSS, Shadcn UI, Web3.js.
- **Backend Development:** Node.js, Express.js, Payload CMS, Django, REST APIs, MongoDB, MySQL, IPFS.
- **Cloud & DevOps:** CI/CD Pipelines, Docker, AWS.
- **Tools & Libraries:** Git, Jest, Swagger & Bruno API, ChatGPT, Pinecone Assistant, HD(Hierarchical Deterministic) Keys.

WORK EXPERIENCE

Clar Technologies, India

Software Development Engineer • Coimbatore, India

July 2024 - Present

Software Engineer Intern • Coimbatore, India

July 2023 - June 2024

- Designed responsive interfaces using **React TypeScript**, integrating **Express.js** backend and **Cosmos SDK**, achieving a 30% improvement in user interaction efficiency.
- Authored API documentation with Swagger and Bruno, improving backend API testing and validation while reducing documentation time by 30%.
- Streamlined pull request reviews with Bitbucket pipelines, increasing development efficiency and achieving pipeline integration for 70% of repositories.
- Achieved a secure **decentralized key management system** utilizing HD keys, enhancing system security for Vault Service by 40%.
- Architected a scalable **provenance verifier using the Observable framework and IPFS**, enabling secure and real-time asset validation while increasing user trust by 20%.
- Engineered a robust AWS-based system leveraging **Lambda functions, API Gateways, IAM roles, and Secrets Manager** to streamline and secure key management operations achieving 90% less user key management.
- Optimized backend signing processes with AWS-based solutions and configured API Gateway for efficient SDK-Lambda communication, improving sign performance by 20%.
- Developed a **RAG-based chatbot** leveraging Pinecone Assistant with custom LLM contexts, providing accurate, context-specific responses aligned with 100% organizational accuracy.
- Engineered a **developer-focused chatbot UI** using **Next.js** and **Pinecone Assistant**, simplifying product navigation and improving resource accessibility for external developers by 60%.

PROJECTS

JettonForge: Seamless Token Solutions on TON Blockchain (December 2024 - January 2025)

- Deployed custom Jetton token managing multiple transactions with 99.9% success rate.
- Integrated wallet connections enabling 50+ secure minting and transfer operations.
- Achieved 45% faster transaction processing using optimized TON tools.
- **Tools Used: Typescript, Tact.**

Lask Bank: Redefining Digital Finance (August 2024 - September 2024)

- Built online banking platform serving multiple users with 99.99% user friendly.
- Integrated Plaid enabling secure connections to 5+ financial institutions.
- Improved real-time banking status system downtime by 40%.
- **Tools Used: Next.js, React.js, Typescript, Tailwind CSS, Shadcn UI, Plaid, Chart.js, Sentry.**

EDUCATION

Bachelor of Engineering in Information Science and Engineering | Kumaraguru College of Technology, Coimbatore, India

|September 2020 - May 2024

CGPA: 8.5/10.

CERTIFICATIONS

- Google Data Analytics Professional Certificate (2022) - Google.